RCA SERVICE DESIGN IRP · FINAL PRESENTATION V1

Reading Disagreement

A lightweight service layer for AI-human disagreement in breast screening.

> The critical question is not only whether AI is accurate. It is what the service does when AI output and human judgement do not agree.

Mammogram case → Human judgement + AI output → Disagreement

PROJECT JOURNEY

From broad AI trust to one high-stakes decision moment

Unit 3

Trust and responsibility in healthcare AI

The project began by asking how responsibility is organised when AI becomes part of high-stakes healthcare decisions.

Unit 4

AI-human disagreement in breast screening

The focus narrowed to how radiologists handle disagreement between human judgement and AI output.

Supported by: Unit 3 submission, Current Project Context, Tutorial 1-5 summaries.

CONTEXT

Breast screening is a decision pathway, not a single AI decision

For a non-clinical audience, the key idea is simple: cases are read, compared, sometimes reviewed, then recorded and learned from.

Mammogram acquired → Case reading → Comparison / review → Recall or no recall → Audit / learning

- AI can enter the pathway in different places.
- The final decision remains human-led and organisationally accountable.
- The design opportunity sits around review, record and learning.

RESEARCH INPUTS

The project is built from workflow evidence, expert input and prototype testing

Literature

AI screening studies, pilot implementation notes, arbitration and disagreement scenarios.

Workflow comparison

AI placement models and where disagreement appears across reading pathways.

Interviews

Hutan Ashrafian, Deborah Cunningham, Sophia Zackrisson and clinical-facing expert feedback.

Tutorials

Shift from universal model to anchor model, then from dashboard to service package.

Prototype testing

Soft cue, comparison review, warm handover, decision receipt and outcome-linked learning.

Keep this slide short in the talk: it proves the source base, then quickly moves to findings.

RADIOLOGIST PAIN POINTS

Disagreement is rarely just “AI right, human wrong”

Senior breast reader

Fast, focused, accountable, working inside existing reading and consensus workflows.

Design principle: support judgement without adding heavy steps.

AI signal needs context

A high AI score may mean different things after prior image comparison.

Every extra click matters

High-throughput reading makes heavy forms or repeated prompts risky.

Accountability stays human

AI contributes evidence, but final recall decisions remain human-led.

Learning takes time

Radiologists need to learn AI reading characteristics across cases, not only in one moment.

JOURNEY BREAKDOWN

Where the current disagreement journey becomes fragile

Read

Human judgement is formed under time pressure and professional responsibility.

Mismatch

AI output differs, but the meaning of the mismatch may be unclear.

Review

Evidence, prior images and AI confidence need to be compared without biasing judgement.

Decision

Arbitration resolves the case, but rationale may be weakly captured.

Afterwards

Useful disagreement patterns may not become shared learning material.

PROBLEM / HMW

Current AI-assisted workflows can surface disagreement, but not always support what happens next

The gap is not only detection. It is how teams review, record and learn from AI-human disagreement.

HOW MIGHT WE

help breast screening teams detect, resolve and learn from AI-human disagreements in a way that is accountable, transparent and consistent?

ANALYSIS MODEL

Universal Disagreement Pattern Map, not a universal NHS workflow

Different AI placements change where disagreement appears, who sees it and what handling mechanism is needed.

1

AI before reading

Risk: automation bias or influenced independent judgement.

2

AI as second reader

Risk: unclear override rationale and escalation criteria.

3

AI as safety net

Risk: late second-look trigger and uncertainty about reopening decisions.

4

AI as triage

Risk: routing boundary decisions and exception handling.

DESIGN ANCHOR

Specific Anchor Model: AI as second reader + human-led arbitration

This anchor makes comparison, mismatch, escalation, rationale and accountability visible.

Human judgement
Independent reader interpretation

AI output
Risk score, flag or evidence layer

Human-AI comparison
Disagreement detected and classified

Consensus / arbitration
Human-led senior review

Record + learning
Decision rationale becomes reusable evidence

PROPOSAL

A three-level disagreement support service

The proposal is not a dashboard. It is a lightweight service layer across case, team and system learning levels.

Case level

Help the reader inspect the disagreement without turning AI into an instruction.

Soft signal · Comparison review · Prior image cue

Team level

Help a difficult case move into peer or senior review with context intact.

Peer pause · Warm handover · Arbitration support

System level

Help selected difficult cases become non-blame learning and calibration material.

Decision receipt · Learning card · Review loop

SERVICE BLUEPRINT

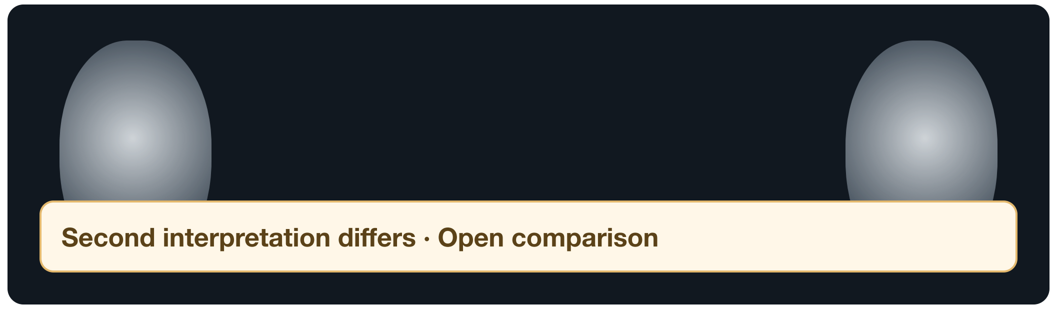
The Disagreement Care Loop connects touchpoints, roles and records

Frontstage	Soft disagreement signal	Comparison review	Peer pause	Warm handover	Human-led review	Learning discussion
Backstage	AI output and human judgement compared	Evidence and prior-image status prepared	Escalation route selected	Context travels with case	Rationale auto-captured	Outcome-linked case selection
Record	Mismatch type	Evidence checked	Support requested	Handover summary	Non-blame receipt	Calibration case card

Service logic: Detect → Resolve → Learn. The final decision is not the end of the service; it is the gateway to learning.

PROTOTYPE + TESTING

Three prototype moments, refined by expert feedback



1. Soft signal

A calm cue makes disagreement visible without acting like an alarm.

2. Warm handover

A short, mostly auto-filled summary carries context into consensus or senior review.

3. Decision receipt

The service preserves rationale now, then creates learning material only after outcome is known.